

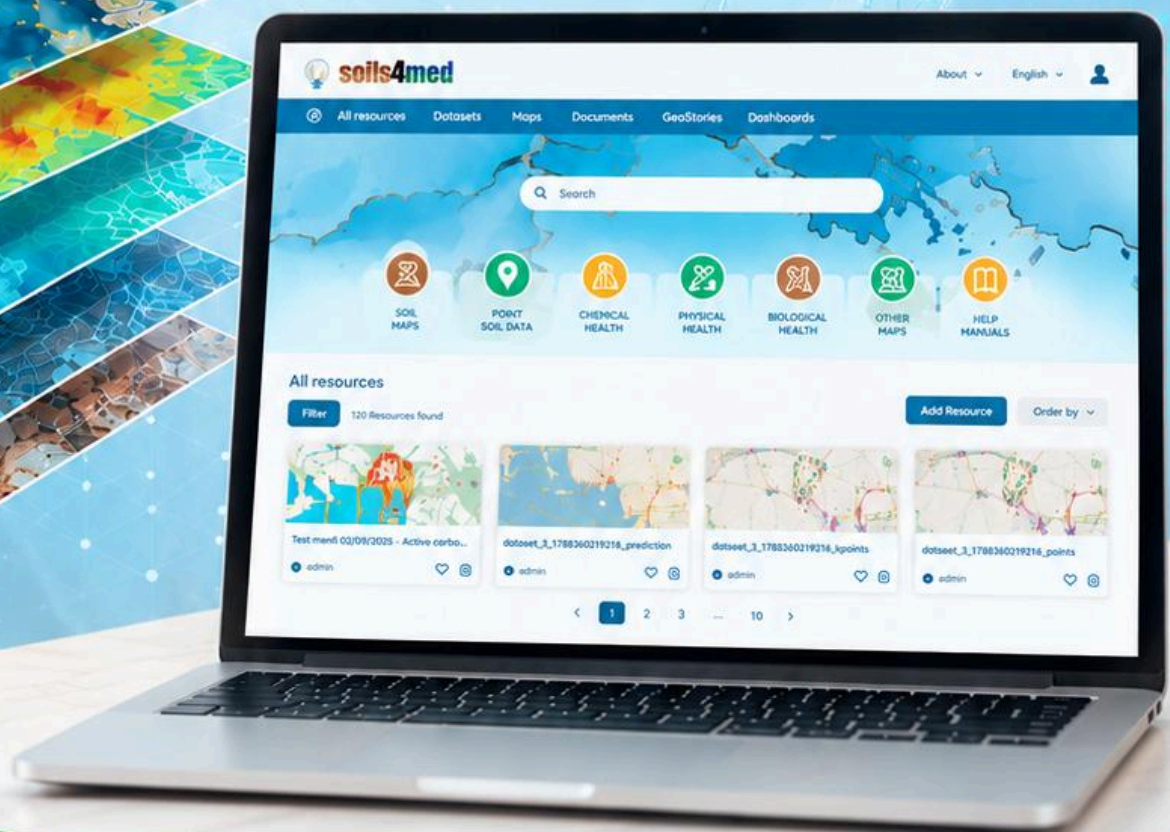
soils4med

SOIL health monitoring and information systems FOR
sustainable soil management in the MEDiterranean region

MedSIS manual

Point Soil Data

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DATASETS



MAPS



DOCUMENTS



GEOSTORIES



DASHBOARDS

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Delivery dates: **10 Settembre 2026**

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Overview

The following section provides a list of the GIS layers containing Soil Point Data available in Soils4Med for the characterization and description of Soil Samples and Soil Profiles.

The structure and attributes of these layers are based on the World Reference Base for Soil Resources ([WRB 2022](#)) standard and on the data structure defined for the S4M soil data spreadsheets (XLSX) used for soil data uploads described in two files located in the MED-SIS repository on GitHub:

- [MED-SIS-TEMPLATE_POINT_SOIL_DATA.xlsx](#) : XLSx template
- [MED_SIS-Manual_Soil_Data_Template_Guidelines.pdf](#) : Editor Guideline

The current legend is built from a [SLD](#) file:

```
<?xml version="1.0" encoding="UTF-8"?>
<StyledLayerDescriptor version="1.0.0"
  xsi:schemaLocation="http://www.opengis.net/sld StyledLayerDescriptor.xsd"
  xmlns="http://www.opengis.net/sld" xmlns:ogc="http://www.opengis.net/ogc"
  xmlns:xlink="http://www.w3.org/1999/xlink" xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance">
  <NamedLayer>
    <Name>point_soil_data</Name>
    <UserStyle>
      <Title>Point Soil Data</Title>
      <FeatureTypeStyle>
        <Rule>
          <Title>Point</Title>
          <PointSymbolizer>
            <Graphic>
              <Mark>
                <WellKnownName>circle</WellKnownName>
                <Fill> <CssParameter name="fill">#FFAA00</CssParameter> </Fill>
                <Stroke>
                  <CssParameter name="stroke">#990000</CssParameter>
                  <CssParameter name="stroke-width">2</CssParameter>
                </Stroke>
              </Mark>
            <Size>8</Size>
          </Graphic>
        </PointSymbolizer>
      </Rule>
    </FeatureTypeStyle>
  </UserStyle>
</NamedLayer>
</StyledLayerDescriptor>
```

The SQL used to create Database View is located in a Django migration file:
(https://github.com/crs4/med_sis/blob/main/src/backoffice/migrations/0003_sqlview_sections.py)

Point Soil Data - How To access to data

To access the soil data, the user must be registered in the system and submit a request to the data managers—specifically, the SIS staff—using the contact details provided on the "Data Policy" page. The link to access the "Data Policy" page is located at the bottom of the home page, as shown in Figure 1

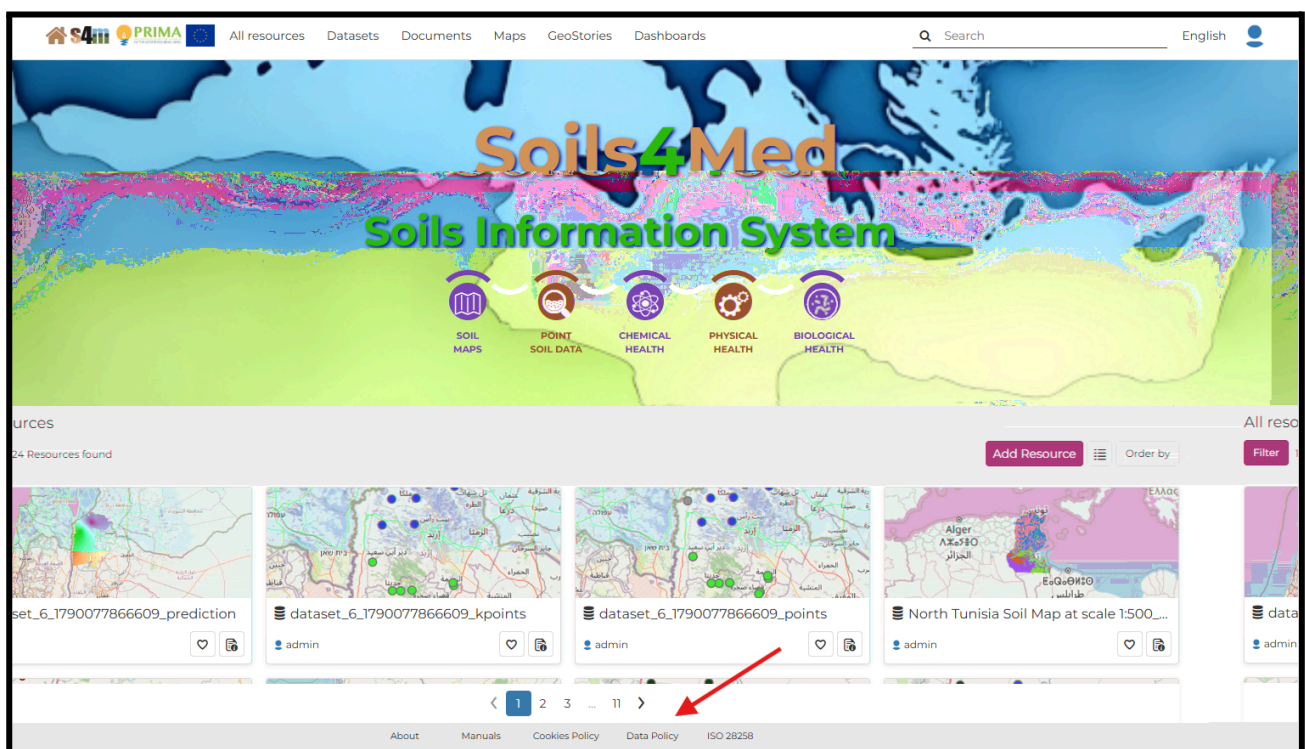


Figura 1

The request must contain :

- A list of **SIS code names** of point-based soil data sections
- The **area of Interest** (a polygon or an administrative division name's)

The request may also include:

- The **period of observations** (start and end dates) or all dates
- The **depth** (upper and lower depth) or all depths
- The **type** : *Auger hole*, *Minipit* or *Profile* or all types

- The Identifier or name for the **project**, or survey activity/campaign, in the frame of which the points soil data data were generated or all projects

- The **method** used to carry out the field survey or all methods:

Identifier	Code	Description
SURVEY_METHODS:CP	CP	method: CPCS;
SURVEY_METHODS:FA	FA	method: FAO
SURVEY_METHODS:FA06	FA06	method: FAO; year: 2006
SURVEY_METHODS:FA52	FA52	method: FAO; year: 1952
SURVEY_METHODS:FA77	FA77	method: FAO; year: 1977
SURVEY_METHODS:FA90	FA90	method: FAO; year: 1990
SURVEY_METHODS:US	US	method: USDA soil survey manual
SURVEY_METHODS:US18	US18	method: USDA soil survey manual ; year: 2018
SURVEY_METHODS:US52	US52	method: USDA soil survey manual ; year: 1952
SURVEY_METHODS:US62	US62	method: USDA soil survey manual ; year: 1962
SURVEY_METHODS:US75	US75	method: USDA soil survey manual ; year: 1975
SURVEY_METHODS:US93	US93	method: USDA soil survey manual ; year: 1993
SURVEY_METHODS:WR	WR	method: WRB;
SURVEY_METHODS:WR22	WR22	method: WRB; year: 2022

After creation and if the data managers evaluation is ok, the dataset with filtered points will be available for view and download in the catalogue

General and Surfaces Sections

Point General Data

General Info Section for Soil Point Data

Legend:



SIS code name: point_general_geo

Short abstract: General Info Section for Soil Point Data

Category: Point Soil Data

Dataset table fields:

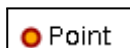
name	description
<u>id</u>	Identifier of the record and Identifier of the point soil data
<u>date</u>	Survey date associated with the point
<u>surveyors</u>	Names of describing authors
<u>location</u>	Location name
<u>lat_wgs84</u>	Latitude in WGS84 SYSTEM
<u>lon_wgs84</u>	Longitude in WGS84 SYSTEM
<u>gps</u>	Coordinates from GPS or not
<u>elev_m_asl</u>	Elevation above sea level [m]
<u>elev_dem</u>	Elevation from DEM a.s.l. [m]
<u>point_type</u>	Point Soil Data type See taxonomy "POINTS_DATA_TYPES" in the manual "S4M Taxonomies"
<u>survey_m_id</u>	Method used to carry out the field survey in order to allow correct interpretation of the information entered See taxonomy "SURVEY_METHODS" in the manual "S4M Taxonomies"

<u>project</u>	Identifier or name for the project, or survey activity/campaign, in the frame of which the point soil data data were generated
<u>type_id</u>	Point Soil Data type See taxonomy "POINTS_DATA_TYPES" in the manual "S4M Taxonomies"
<u>old_cls</u>	Old classification
<u>old_id</u>	Original identifier
<u>new_cls</u>	New classification
<u>cls_sys_id</u>	Classification system See taxonomy "CLASSIFICATION_SYSTEMS" in the manual "S4M Taxonomies"
<u>notes</u>	Generic field added to insert comments
<u>climateandweather_id</u>	Identifier used for linking to section ClimateAndWeather
<u>landformtopography_id</u>	Identifier used for linking to section LandformTopography
<u>landuse_id</u>	Identifier used for linking to section LandUse
<u>surface_id</u>	Identifier used for linking to section Surface
<u>surfaceunevenness_id</u>	Identifier used for linking to section SurfaceUnevenness
<u>n_layer,</u>	Number of layers
<u>horizon_designation</u>	Soil Horizon Designations

Landform and Topography

Landform Topography Info Section for Soil Point Data

Legend:



SIS code name: landform_topography_geo

Short abstract: Landform Topography Info Section for Soil Point Data

Category: Point Soil Data

Indicator table fields:

name	description
<u>id</u>	Identifier of the record and Identifier of the point soil data
<u>point_id</u>	Identifier of the point in Point General Data
<u>point_type</u>	Point Soil Data type See taxonomy "POINTS_DATA_TYPES" in the manual "S4M Taxonomies"
<u>date</u>	Survey date associated with the point
<u>project</u>	Identifier or name for the project, or survey activity/campaign, in the frame of which the point soil data data were generated
<u>survey_m_id</u>	Method used to carry out the field survey in order to allow correct interpretation of the information entered See taxonomy "SURVEY_METHODS" in the manual "S4M Taxonomies"
<u>grad_ups</u>	Gradient upslope [%]
<u>grad_downs</u>	Gradient downslope [%]
<u>slope_asp</u>	Slope aspect [°]
<u>slope_shp_id</u>	Slope shape type See taxonomy "SHAPES" in the manual "S4MTaxonomies"
<u>position_id</u>	Profile position See taxonomy "POSITION" in the manual "S4M Taxonomies"
<u>landform1_id</u>	Geomorphic features: dominant land form See taxonomy "LANDFORMS" in the manual "S4M Taxonomies"
<u>activity1_id</u>	Geomorphic features: activity in the dominant land form See taxonomy "ACTIVITIES" in the manual "S4M Taxonomies"
<u>landform2_id</u>	Geomorphic features: secondary land form

	See taxonomy "LANDFORMS" in the manual "S4M Taxonomies"
<u>activity2_id</u>	Geomorphic features: activity in the secondary land form See taxonomy "ACTIVITIES" in the manual "S4M Taxonomies"
<u>geo_descr</u>	Geomorphic features: Description

Climate and Weather

Climate and Weather Info Section for Soil Point Data

Legend:



SIS code name: Climate and Weather

Short abstract: Climate and Weather Info Section for Soil Point Data

Category: Point Soil Data

Involved SIS tables: climate_and_weather_geo

Indicator table fields:

name	description
<u>id</u>	Identifier of the record and Identifier of the point soil data
<u>point_id</u>	Identifier of the point in Point General Data
<u>point_type</u>	Point Soil Data type See taxonomy "POINTS_DATA_TYPES" in the manual "S4M Taxonomies"
<u>date</u>	Survey date associated with the point
<u>project</u>	Identifier or name for the project, or survey activity/campaign, in the frame of which the point soil data data were generated
<u>survey_m_id</u>	Method used to carry out the field survey in order to allow correct interpretation of the information entered See taxonomy "SURVEY_METHODS" in the manual "S4M Taxonomies"
<u>clim_koppen</u>	Climate according to Köppen (1936) See taxonomy "CLIMATE_KOPPEN" in the manual "S4M Taxonomies"
<u>eco_shultz</u>	Ecozone according to Schultz (2005 adapted) See taxonomy "ECOZONE_SHULTZ" in the manual "S4M Taxonomies"
<u>season</u>	Season of description See taxonomy "SEASON" in the manual "S4M Taxonomies"
<u>curr_weath</u>	Current weather conditions See taxonomy "CURRENT_WEATHER" in the manual "S4M Taxonomies"

<u>past_weath</u>	Past weather conditions See taxonomy "PAST_WEATHER" in the manual "S4M Taxonomies"
<u>soil_temp</u>	Soil temperature regime See taxonomy "SOIL_TEMP_REGIME" in the manual "S4M Taxonomies"
<u>soil_moist</u>	Soil moisture regime See taxonomy "SOIL_MOIST_REGIME" in the manual "S4M Taxonomies"

Surface land use

Surface Land Use Info Section for Soil Point Data

Legend



SIS code name: land_use_geo

Short abstract: Surface Land Use Info Section for Soil Point Data

Category: Point Soil Data

Indicator table fields:

name	description
<u>id</u>	Identifier of the record and Identifier of the point soil data
<u>point_id</u>	Identifier of the point in Point General Data
<u>point_type</u>	Point Soil Data type See taxonomy "POINTS_DATA_TYPES" in the manual "S4M Taxonomies"
<u>date</u>	Survey date associated with the point
<u>project</u>	Identifier or name for the project, or survey activity/campaign, in the frame of which the point soil data data were generated
<u>survey_m_id</u>	Method used to carry out the field survey in order to allow correct interpretation of the information entered See taxonomy "SURVEY_METHODS" in the manual "S4M Taxonomies"
<u>use_id</u>	Land Use

	See taxonomy "USES" in the manual "S4M Taxonomies"
<u>corine_id</u>	Corine class See taxonomy "CORINE" in the manual "S4M Taxonomies"
<u>nc_gs_veget1_id</u>	Not Cultivated Vegetation type 1.1 (dominant) in the Ground stratum See taxonomy "VEGETATION_TYPES" in the manual "S4M Taxonomies"
<u>nc_gs_veget2_id</u>	Not Cultivated Vegetation type 1.2 in the Ground stratum See taxonomy "VEGETATION_TYPES" in the manual "S4M Taxonomies"
<u>nc_gs_veget3_id</u>	Not Cultivated Vegetation type 1.3 in the Ground stratum See taxonomy "VEGETATION_TYPES" in the manual "S4M Taxonomies"
<u>nc_gs_avg_height</u>	Not Cultivated Average height [m] in the Ground stratum
<u>nc_gs_max_height</u>	Not Cultivated Maximum height [m] in the Ground stratum
<u>nc_gs_area</u>	Not Cultivated Vegetation cover by area of the ground cover [%] in the Ground stratum
<u>nc_gs_species1</u>	Not Cultivated Species 1 in the Ground stratum
<u>nc_gs_species2</u>	Not Cultivated Species 2 in the Ground stratum
<u>nc_gs_species3</u>	Not Cultivated Species 3 in the Ground stratum
<u>nc_us_veget1_id</u>	Not Cultivated Vegetation type 2.1 (dominant) in the Upper stratum See taxonomy "VEGETATION_TYPES" in the manual "S4M Taxonomies"
<u>nc_us_veget2_id</u>	Not Cultivated Vegetation type 2.2 in the Upper stratum See taxonomy "VEGETATION_TYPES" in the manual "S4M Taxonomies"
<u>nc_us_veget3_id</u>	Not Cultivated Vegetation type 2.3 in the Upper stratum See taxonomy "VEGETATION_TYPES" in the manual "S4M Taxonomies"
<u>nc_us_avg_height</u>	Not Cultivated Average height [m] in the Upper stratum
<u>nc_us_max_height</u>	Not Cultivated Maximum height [m] in the Upper stratum
<u>nc_us_area</u>	Not Cultivated Vegetation cover by area of the ground cover [%] in the Upper stratum
<u>nc_us_species1</u>	Not Cultivated Species 1 in the Upper stratum

<u>nc_us_species2</u>	Not Cultivated Species 2 in the Upper stratum
<u>nc_us_species3</u>	Not Cultivated Species 3 in the Upper stratum
<u>nc_ms_veget1_id</u>	Not Cultivated Vegetation type 3.1 (dominant) in the Middle-stratum See taxonomy "VEGETATION_TYPES" in the manual "S4M Taxonomies"
<u>nc_ms_veget2_id</u>	Not Cultivated Vegetation type 3.2 in the Middle-stratum See taxonomy "VEGETATION_TYPES" in the manual "S4M Taxonomies"
<u>nc_ms_veget3_id</u>	Not Cultivated Vegetation type 3.3 in the Middle-stratum See taxonomy "VEGETATION_TYPES" in the manual "S4M Taxonomies"
<u>nc_ms_avg_height</u>	Not Cultivated Average height [m] in the Middle-stratum
<u>nc_ms_max_height</u>	Not Cultivated Maximum height [m] in the Middle-stratum
<u>nc_ms_area</u>	Not Cultivated Vegetation cover by area of the ground cover [%] in the Middle-stratum
<u>nc_ms_species1</u>	Not Cultivated Species 1 in the Middle-stratum
<u>nc_ms_species2</u>	Not Cultivated Species 2 in the Middle-stratum
<u>nc_ms_species3</u>	Not Cultivated Species 3 in the Middle-stratum
<u>cult_type_id</u>	Cultivation type See taxonomy "CULTIVATION_TYPES" in the manual "S4M Taxonomies"
<u>cult_area</u>	Field added to Vegetation cover [%]
<u>cult_actual1</u>	Actual cultivated species 1 (dominant)
<u>cult_actual2</u>	Actual cultivated species 2
<u>cult_actual3</u>	Actual cultivated species 3
<u>cult_last1</u>	Last cultivated species 1 (dominant)
<u>cult_last2</u>	Last cultivated species 2
<u>cult_last3</u>	Last cultivated species 3
<u>cult_cessation</u>	Month and year of harvest or cultivation cessation of Last cultivated species 1 (dominant)
<u>cult_rotation1</u>	Rotational cultivated species 1 (most frequent)

<u>cult_rotation2</u>	Rotational cultivated species 2
<u>cult_rotation3</u>	Rotational cultivated species 3
<u>cult_prod1_tech_id</u>	First special technique to enhance site productivity (dominant) See taxonomy "PROD_TECHNIQUES" in the manual "S4M Taxonomies"
<u>cult_prod2_tech_id</u>	Second special technique to enhance site productivity See taxonomy "PROD_TECHNIQUES" in the manual "S4M Taxonomies"
<u>cult_prod3_tech_id</u>	Third special technique to enhance site productivity See taxonomy "PROD_TECHNIQUES" in the manual "S4M Taxonomies"

Surface general

Surface general info section for Soil Point Data

Legend



SIS code name: surface_geo

Short abstract: Surface general info section for Soil Point Data

Category: Point Soil Data

Indicator table fields:

name	description
<u>id</u>	Identifier of the record and Identifier of the point soil data
<u>point_id</u>	Identifier of the point in Point General Data
<u>point_type</u>	Point Soil Data type See taxonomy "POINTS_DATA_TYPES" in the manual "S4M Taxonomies"
<u>date</u>	Survey date associated with the point

<u>project</u>	Identifier or name for the project, or survey activity/campaign, in the frame of which the point soil data data were generated
<u>survey_m_id</u>	Method used to carry out the field survey in order to allow correct interpretation of the information entered See taxonomy "SURVEY_METHODS" in the manual "S4M Taxonomies"
<u>litter_area</u>	Litter Layer Area covered [%]
<u>litter_avg_thick</u>	Litter Layer Average thickness [cm]
<u>litter_max_thick</u>	Litter Layer Maximum thickness [cm]
<u>bedr_form</u>	Bedrock Formation name
<u>bedr_lith_id</u>	Surface Bedrock Lithology See taxonomy "PARENT_MATERIALS" in the manual "S4M Taxonomies"
<u>outc_area</u>	Rock outcrops Area covered [%]
<u>outc_dist</u>	Rock outcrops Average distance [m]
<u>outc_size</u>	Rock outcrops Size [m]
<u>cs_total_area</u>	Coarse Fragments Total area covered [%]
<u>cs_class1size_id</u>	Coarse Fragments Size class 1 (dominant) See taxonomy "COARSE_SIZES" in the manual "S4M Taxonomies"
<u>cs_class2size_id</u>	Coarse Fragments Size class 2 See taxonomy "COARSE_SIZES" in the manual "S4M Taxonomies"
<u>cs_class3size_id</u>	Coarse Fragments Size class 3 See taxonomy "COARSE_SIZES" in the manual "S4M Taxonomies"
<u>cs_class1area</u>	Coarse Fragments Area covered by size class 1 [%]
<u>cs_class2area</u>	Coarse Fragments Area covered by size class 2 [%]
<u>cs_class3area</u>	Coarse Fragments Area covered by size class 3 [%]
<u>desert_ven</u>	Desert features: Ventifacts [%]
<u>desert_var</u>	Desert features: Varnish [%]
<u>ground_form_id</u>	Patterned ground form Class

	See taxonomy "GROUND_FORMS" in the manual "S4M Taxonomies"
<u>crust_area</u>	Surface crusts area covered [%] Surface cracks Width class (dominant)
<u>cracks_width1_id</u>	See taxonomy "WIDTH_TYPES" in the manual "S4M Taxonomies" Distance between surface cracks (dominant)
<u>cracks_dist1_id</u>	See taxonomy "DISTANCE_TYPES" in the manual "S4M Taxonomies" Spatial arrangement of surface cracks (dominant)
<u>cracks_spat_arr1_id</u>	See taxonomy "ARRANGEMENT" in the manual "S4M Taxonomies" Persistence of surface cracks (dominant)
<u>cracks_persist1_id</u>	See taxonomy "SURFACE_CRACKS_PERSISTENCE" in the manual "S4M Taxonomies" Surface cracks Width class (2nd)
<u>cracks_width2_id</u>	See taxonomy "WIDTH_TYPES" in the manual "S4M Taxonomies" Distance between surface cracks (2nd)
<u>cracks_dist2_id</u>	See taxonomy "DISTANCE_TYPES" in the manual "S4M Taxonomies" Spatial arrangement of surface cracks (2nd)
<u>cracks_spat_arr2_id</u>	See taxonomy "ARRANGEMENT" in the manual "S4M Taxonomies" Persistence of surface cracks (2nd)
<u>cracks_persist2_id</u>	See taxonomy "SURFACE_CRACKS_PERSISTENCE" in the manual "S4M Taxonomies" Presence of water
<u>wat_above</u>	See taxonomy "WATER_ABOVE" in the manual "S4M Taxonomies" Drainage conditions
<u>wat_drain</u>	See taxonomy "DRAINAGE_CONDITIONS" in the manual "S4M Taxonomies"
<u>ground_wat</u>	Depth of ground water [cm]
<u>wat_repell</u>	Water repellence

	See taxonomy "WATER_REPELLENCE" in the manual "S4M Taxonomies"
<u>tech_alter</u>	Technical surface alterations See taxonomy "TECH_ALTERATIONS" in the manual "S4M Taxonomies"
<u>debris_type_id</u>	Technogenic debris cover - type of artefact TECHNOGENIC_DEBRIS_COVER_TYPES" in the manual "S4M Taxonomies"
<u>debris_cover</u>	Technogenic debris cover - surface cover
<u>natural_alterations</u>	Natural surface alterations See taxonomy "NATURAL_SURFACE_ALTERATIONS_TYPES" in the manual "S4M Taxonomies"
<u>pat_comp_type</u>	Patterned compaction type See taxonomy "PATTERNED_COMPACTION_TYPES" in the manual "S4M Taxonomies"
<u>pat_comp_cover</u>	Patterned compaction - surface cover
<u>saline_efflorescence</u>	Saline efflorescence on soil surface See taxonomy "CONTINUITY_SALINE_EFFLLORESCENCE_TYPES" in the manual "S4M Taxonomies"
<u>worm_cast</u>	Worm cast abundance XXXnon abundance
<u>lp_signs_ploughing_id</u>	Land preparation - signs of ploughing See taxonomy "SIGN_OF_PLOUGHING_TYPES" in the manual "S4M Taxonomies"
<u>lp_direction_ploughing</u>	Land preparation - direction of ploughing See taxonomy "DIRECTION_OF_PLOUGHING_TYPES" in the manual "S4M Taxonomies"
<u>lp_tillage_mode</u>	Land preparation - Tillage mode See taxonomy "TILLAGE_MODE" in the manual "S4M Taxonomies"
<u>use_of_inputs</u>	Use of inputs See taxonomy "USE_OF_INPUT" in the manual "S4M Taxonomies"
<u>irrigation</u>	Irrigation Type

	See taxonomy "IRRIGATION_TYPES" in the manual "S4M Taxonomies"
<u>conservation</u>	Type of conservation measure See taxonomy "CONSERVATION_MEASURE_TYPES" in the manual "S4M Taxonomies"

Surface Unevenness

Surface unevenness info section for Soil Point Data

Legend



SIS code name: surface_unevenness_geo

Short abstract: Surface unevenness info section for Soil Point Data

Category: Point Soil Data

Indicator table fields:

name	description\
<u>id</u>	Identifier of the record and Identifier of the point soil data
<u>point_id</u>	Identifier of the point in Point General Data
<u>point_type</u>	Point Soil Data type See taxonomy "POINTS_DATA_TYPES" in the manual "S4M Taxonomies"
<u>date</u>	Survey date associated with the point
<u>project</u>	Identifier or name for the project, or survey activity/campaign, in the frame of which the point soil data data were generated
<u>survey_m_id</u>	Method used to carry out the field survey in order to allow correct interpretation of the information entered See taxonomy "SURVEY_METHODS" in the manual "S4M Taxonomies"
<u>nat_type_id</u>	Natural Type See taxonomy "NATURAL_TYPES" in the manual "S4M Taxonomies"

<u>nat_avg_h</u>	Average height difference [m]
<u>nat_elev</u>	Average diameter of the elevated areas [m]
<u>nat_dist</u>	Average distance between the height maxima [m]
<u>hum_type1_id</u>	Human made dominant type See taxonomy "HUMAN_MADE_TYPES" in the manual "S4M Taxonomies"
<u>hum_type2_id</u>	Human made secondary type See taxonomy "HUMAN_MADE_TYPES" in the manual "S4M Taxonomies"
<u>hum_noter_h</u>	For terraces Average height [cm]
<u>hum_noter_h</u>	For all other features: average difference of highest and lowest points [cm]
<u>hum_noter_w</u>	For all other features: average width/ length [cm]
<u>hum_noter_d</u>	For all other features: average distance between depth/height maxima [cm]
<u>ero_type_id</u>	Surface unevenness erosion type See taxonomy "EROSION_TYPES" in the manual "S4M Taxonomies"
<u>ero_area</u>	Total area affected by erosion [%]
<u>ero_degree_id</u>	Surface unevenness erosion degree See taxonomy "EROSION_DEGREE" in the manual "S4M Taxonomies"
<u>ero_activ_id</u>	Type of surface unevenness erosion activity See taxonomy "EROSION_ACTIVITY" in the manual "S4M Taxonomies"
<u>position_id</u>	Position of the soil profile See taxonomy "PROFILE_POSITION" in the manual "S4M Taxonomies"

Layer Description Sections

Layer General Data

Point layers Info Section for Soil Point Data

Legend

SIS code name: point_layer_geo

Short abstract: Point layers Info Section for Soil Point Data

Category: Point Soil Data

Indicator table fields:

name	description
id	Identifier of the record and Identifier of the point soil data
point_id	Identifier of the point in Point General Data
point_type	Point Soil Data type See taxonomy "POINTS_DATA_TYPES" in the manual "S4M Taxonomies"
date	yyyy-mm-dd ISO 8601 Date

<u>lower_bound</u>	Layer lower boundary
<u>thickness</u>	Layer thickness must be layer.lower minus layer.upper
<u>hom_part</u>	Described layer parts by exposed area [%] (if all: write 100)
<u>hom_alluvt_id</u>	Layer composed of several strata of alluvial sediments or of tephra, see table 8.26 WRB 4th edition 2022 See taxonomy "ALLUVIAL_TEPHRA" in the manual "S4M Taxonomies"
<u>wat_satur_id</u>	Water saturation see table 8.27 WRB 4th edition 2022 See taxonomy "WATER_SATURATION" in the manual "S4M Taxonomies"
<u>wat_status_id</u>	Soil water status see table 8.28 WRB 4th edition 2022 See taxonomy "WATER_STATUS" in the manual "S4M Taxonomies"
<u>o_mineral_id</u>	Organo technic or mineral layer, see table 8.29 WRB 4th edition 2022. See taxonomy "ORGANIC_MINERAL_TYPES" in the manual "S4M Taxonomies"
<u>bounddist_id</u>	Distinctness of the layers lower boundary, see table 8.30 WRB 4th edition 2022 See taxonomy "BOUNDARIES" in the manual "S4M Taxonomies"
<u>boundshape_id</u>	Boundary shape. The characteristic refers to the lower boundary or if the shape is broken to the entire layer see table 8.31 WRB 4th edition 2022 See taxonomy "L_SLOPE_SHAPE" in the manual "S4M Taxonomies"
<u>wind_id</u>	Evidence of wind deposition, see table 8.32 WRB 4th edition 2022 See taxonomy "WIND" in the manual "S4M Taxonomies"
<u>tex_cls_id</u>	Texture class, see table 8.39 WRB 4th edition 2022 See taxonomy "TEXTURE_CLASSES" in the manual "S4M Taxonomies"
<u>tex_subcls_id</u>	Subclass of the texture classes sand and loamy sand, see table 8.40 WRB 4th edition 2022 See taxonomy "TEXTURE_SUBCLASSES" in the manual "S4M Taxonomies"

struct_w_s

Wedge-shaped aggregates tilted between $\geq 10^\circ$ and $\leq 60^\circ$ from the horizontal: abundance by volume [%]

stress_fts_pressfaces

Abundance of pressure faces [%]

stress_fts_slicksides

Abundance of slickensides [%]

coars_text

Coarser-textured parts of any orientation and a width of ≥ 0.5 cm by exposed area [%]

coars_text_v_tongues

Continuous vertical tongues of coarser-textured parts with a horizontal extension of ≥ 1 cm by exposed area [%]

coars_text_depth

Depth range where tongues cover $\geq 10\%$ of the exposed area [cm]

coars_text_h_area

Horizontal area covered [%]

rh_value_id

Rh value, see table 8.59 WRB 4th edition 2022

See taxonomy "RH_VALUE" in the manual "S4M taxonomies"

weathering

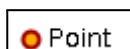
Abundance of Fe ne

	"SEALING_AGENT" in the manual "S4M Taxonomies"
<u>sur_crusts_sealing2_id</u>	Sealing agent 2 of surface crusts, see table 8.78 WRB 4th edition 2022 See taxonomy "SEALING_AGENT" in the manual "S4M Taxonomies"
<u>sur_crusts_sealing3_id</u>	Sealing agent 3 of surface crusts, see table 8.78 WRB 4th edition 2022 See taxonomy "SEALING_AGENT" in the manual "S4M Taxonomies"
<u>sur_crusts_avg_thickn</u>	Average thickness of surface crusts
<u>fracts</u>	Volume occupied by the fractures [%]
<u>avg_fracts</u>	Average distance between the fractures [cm]
<u>volc_abund_id</u>	Abundance of volcanic glasses in the sand and coarse silt fraction, see table 8.79 WRB 4th edition 2022 See taxonomy "ABUNDANCE_PARTICLES_TYPES" in the manual "S4M Taxonomies"
<u>volc_thnaf_id</u>	Thixotropy and NaF field test, see table 8.80 WRB 4th edition 2022 See taxonomy "THIXOTROPY_NAF" in the manual "S4M Taxonomies"
<u>pack_dens_id</u>	Packing density, see table 8.83 WRB 4th edition 2022 PACKING_DENSITIES" in the manual "S4M Taxonomies"
<u>bulk_dens</u>	Bulk density
<u>parent_mat_id</u>	Type of parent material, see table 8.91 WRB 4th edition 2022 See taxonomy "PARENT_MATERIALS" in the manual "S4M Taxonomies"
<u>notes</u>	Note about the horizon

Layer coarse fragments

Layers coarse fragments Info Section for Soil Point Data

Legend



SIS code name: layer_coarse_fragments_geo

Short abstract: Layers coarse fragments Info Section for Soil Point Data

Category: Point Soil Data

Indicator table fields:

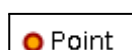
name	description
<u>id</u>	Identifier of the record and Identifier of the point soil data
<u>point_id</u>	Identifier of the point in Point General Data
<u>point_type</u>	Point Soil Data type See taxonomy "POINTS_DATA_TYPES" in the manual "S4M Taxonomies"
<u>date</u>	Survey date associated with the point
<u>project</u>	Identifier or name for the project, or survey activity/campaign, in the frame of which the point soil data data were generated
<u>layer_id</u>	Identifier of the record in Layer General Data
<u>upper</u>	Layer upper depth
<u>lower</u>	Layer lower depth
<u>survey_m_id</u>	Method used to carry out the field survey in order to allow correct interpretation of the information entered See taxonomy "SURVEY_METHODS" in the manual "S4M Taxonomies"
<u>abundance</u>	Total abundance of coarse fragments by volume [%]
<u>size1_id</u>	Size and shape class 1 of the coarse fragments, see table 8.33 WRB 4th edition 2022 modified by inserting 2 values for the shape field: subrounded and flat See taxonomy "SIZE_SHAPE_TYPES" in the manual "S4M Taxonomies"

<u>size2_id</u>	Size and shape class 2 of the coarse fragments, see table 8.33 WRB 4th edition 2022 modified by inserting 2 values for the shape field: subrounded and flat See taxonomy "SIZE_SHAPE_TYPES" in the manual "S4M Taxonomies"
<u>size3_id</u>	Size and shape class 3 of the coarse fragments, see table 8.33 WRB 4th edition 2022 modified by inserting 2 values for the shape field: subrounded and flat See taxonomy "SIZE_SHAPE_TYPES" in the manual "S4M Taxonomies"
<u>size4_id</u>	Size and shape class 4 of the coarse fragments, see table 8.33 WRB 4th edition 2022 modified by inserting 2 values for the shape field: subrounded and flat See taxonomy "SIZE_SHAPE_TYPES" in the manual "S4M Taxonomies"
<u>weath1_id</u>	Weathering stage of size and shape class 1, see table 8.34 WRB 4th edition 2022 See taxonomy "WEATHERING_STAGE" in the manual "S4M Taxonomies"
<u>weath2_id</u>	Weathering stage of size and shape class 2, see table 8.34 WRB 4th edition 2022 See taxonomy "WEATHERING_STAGE" in the manual "S4M Taxonomies"
<u>weath3_id</u>	Weathering stage of size and shape class 3, see table 8.34 WRB 4th edition 2022 See taxonomy "WEATHERING_STAGE" in the manual "S4M Taxonomies"
<u>weath4_id</u>	Weathering stage of size and shape class 4, see table 8.34 WRB 4th edition 2022 See taxonomy "WEATHERING_STAGE" in the manual "S4M Taxonomies"
<u>abundance1</u>	Abundance of size and shape class 1 by volume [%]
<u>abundance2</u>	Abundance of size and shape class 2 by volume [%]
<u>abundance3</u>	Abundance of size and shape class 3 by volume [%]
<u>abundance4</u>	Abundance of size and shape class 4 by volume [%]
<u>litho_type1_id</u>	Lithology of size and shape class 1, see table 8.91 WRB 4th edition 2022 See taxonomy "PARENT_MATERIALS" in the manual "S4M Taxonomies"
<u>litho_type2_id</u>	Lithology of size and shape class 2, see table 8.91 WRB 4th edition 2022 See taxonomy "PARENT_MATERIALS" in the manual "S4M Taxonomies"

	"S4M Taxonomies"
<u>litho_type3_id</u>	Lithology of size and shape class 3, see table 8.91 WRB 4th edition 2022 See taxonomy "PARENT_MATERIALS" in the manual "S4M Taxonomies"
<u>litho_type4_id</u>	Lithology of size and shape class 4, see table 8.91 WRB 4th edition 2022 See taxonomy "PARENT_MATERIALS" in the manual "S4M Taxonomies"
<u>free_pore</u>	Free large pores between coarse fragments by volume [%]

Layer remnants

Legend



SIS code name: layer_remnants_geo

Short abstract: Layers remnants Info Section for Soil Point Data

Category: Point Soil Data

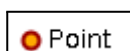
Indicator table fields:

name	description
<u>id</u>	Identifier of the record and Identifier of the point soil data
<u>point_id</u>	Identifier of the point in Point General Data
<u>point_type</u>	Point Soil Data type See taxonomy "POINTS_DATA_TYPES" in the manual "S4M Taxonomies"
<u>date</u>	Survey date associated with the point
<u>project</u>	Identifier or name for the project, or survey activity/campaign, in the frame of which the point soil data data were generated
<u>layer_id</u>	Identifier of the record in Layer General Data

<u>upper</u>	Layer upper depth
<u>lower</u>	Layer lower depth
<u>survey_m_id</u>	Method used to carry out the field survey in order to allow correct interpretation of the information entered See taxonomy "SURVEY_METHODS" in the manual "S4M Taxonomies"
<u>abundance</u>	Total abundance of remnants of broken-up cemented layers by volume [percentage]
<u>cementing1_id</u>	Cementing agent 1 in Remnants of broken-up cemented layers: , see table 8.35 WRB 4th edition 2022 and taxonomy "CEMENTING_AGENTS" in the manual "S4M Taxonomies"
<u>cementing2_id</u>	Cementing agent 2 in Remnants of broken-up cemented layers: , see table 8.35 WRB 4th edition 2022 and taxonomy "CEMENTING_AGENTS" in the manual "S4M Taxonomies"
<u>size1_id</u>	Size and shape class of remnants cemented by cementing agent 1, see table 8.33 WRB 4th edition 2022 modified by inserting 2 values for the shape field: subrounded and flat. See taxonomy "SIZE_SHAPE_TYPES" in the manual "S4M Taxonomies"
<u>size2_id</u>	Size and shape class of remnants cemented by cementing agent 2, see table 8.33 WRB 4th edition 2022 modified by inserting 2 values for the shape field: subrounded and flat See taxonomy "SIZE_SHAPE_TYPES" in the manual "S4M Taxonomies"
<u>abundance1</u>	Abundance of remnants cemented by cementing agent 1 by volume [%]
<u>abundance2</u>	Abundance of remnants cemented by cementing agent 2 by volume [%]

Layer Artefacts

Legend



SIS code name: layer_artefacts_geo

Short abstract: Layers artefacts Info Section for Soil Point Data

Category: Point Soil Data

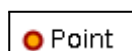
Indicator table fields:

name	description
<u>id</u>	Identifier of the record and Identifier of the point soil data
<u>point_id</u>	Identifier of the point in Point General Data
<u>point_type</u>	Point Soil Data type See taxonomy "POINTS_DATA_TYPES" in the manual "S4M Taxonomies"
<u>date</u>	Survey date associated with the point
<u>project</u>	Identifier or name for the project, or survey activity/campaign, in the frame of which the point soil data data were generated
<u>layer_id</u>	Identifier of the record in Layer General Data
<u>upper</u>	Layer upper depth
<u>lower</u>	Layer lower depth
<u>survey_m_id</u>	Method used to carry out the field survey in order to allow correct interpretation of the information entered See taxonomy "SURVEY_METHODS" in the manual "S4M Taxonomies"
<u>abundance</u>	Total abundance of artefacts by volume [%]
<u>black_carb</u>	Black carbon by exposed area [%]
<u>type1_id</u>	Artefact type 1, see table 8.36 WRB 4th edition 2022 See taxonomy "ARTEFACT_TYPES" in the manual "S4M Taxonomies"

<u>type2_id</u>	Artefact type 2, see table 8.36 WRB 4th edition 2022 See taxonomy "ARTEFACT_TYPES" in the manual "S4M Taxonomies"
<u>type3_id</u>	Artefact type 3, see table 8.36 WRB 4th edition 2022 See taxonomy "ARTEFACT_TYPES" in the manual "S4M Taxonomies"
<u>type4_id</u>	Artefact type 4, see table 8.36 WRB 4th edition 2022 See taxonomy "ARTEFACT_TYPES" in the manual "S4M Taxonomies"
<u>type5_id</u>	Artefact type 5, see table 8.36 WRB 4th edition 2022 See taxonomy "ARTEFACT_TYPES" in the manual "S4M Taxonomies"
<u>size1_id</u>	Artefact type 1 size, see table 8.37 WRB 4th edition 2022 See taxonomy "ARTEFACT_SIZES" in the manual "S4M Taxonomies"
<u>size2_id</u>	Artefact type 2 size, see table 8.37 WRB 4th edition 2022 See taxonomy "ARTEFACT_SIZES" in the manual "S4M Taxonomies"
<u>size3_id</u>	Artefact type 3 size, see table 8.37 WRB 4th edition 2022 See taxonomy "ARTEFACT_SIZES" in the manual "S4M Taxonomies"
<u>size4_id</u>	Artefact type 4 size, see table 8.37 WRB 4th edition 2022 See taxonomy "ARTEFACT_SIZES" in the manual "S4M Taxonomies"
<u>size5_id</u>	Artefact type 5 size, see table 8.37 WRB 4th edition 2022 See taxonomy "ARTEFACT_SIZES" in the manual "S4M Taxonomies"
<u>abundance1</u>	Abundance of artefacts type 1 by volume [%]
<u>abundance2</u>	Abundance of artefacts type 2 by volume [%]
<u>abundance3</u>	Abundance of artefacts type 3 by volume [%]
<u>abundance4</u>	Abundance of artefacts type 4 by volume [%]
<u>abundance5</u>	Abundance of artefacts type 5 by volume [%]

Layer structure

Legend



SIS code name: layer_structure_geo

Short abstract: Layers structures Info Section for Soil Point Data

Category: Point Soil Data

Involved SIS tables: XXXlayer_structures

Indicator table fields:

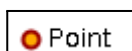
name	description
<u>id</u>	Identifier of the record and Identifier of the point soil data
<u>point_id</u>	Identifier of the point in Point General Data
<u>point_type</u>	Point Soil Data type See taxonomy "POINTS_DATA_TYPES" in the manual "S4M Taxonomies"
<u>date</u>	Survey date associated with the point
<u>project</u>	Identifier or name for the project, or survey activity/campaign, in the frame of which the point soil data data were generated
<u>layer_id</u>	Identifier of the record in Layer General Data
<u>upper</u>	Layer upper depth
<u>lower</u>	Layer lower depth
<u>survey_m_id</u>	Method used to carry out the field survey in order to allow correct interpretation of the information entered. See taxonomy "SURVEY_METHODS" in the manual "S4M Taxonomies"
<u>type_id</u>	Structure Type. See taxonomy "STRUCTURE_TYPES" in the manual "S4M Taxonomies"
<u>abundance_vol</u>	Abundance of structure type 1 by volume [%]
<u>grade_id</u>	Grade of structural units in type 1, see table 8.43 WRB

	4th edition 2022. See taxonomy "STRUCTURE_GRADES" in the manual "S4M Taxonomies"
<u>penetrab_id</u>	Aggregate penetrability for roots in type 1, see table 8.44 WRB 4th edition 2022. See taxonomy "AGGREGATE_PENETRABILITY" in the manual "S4M Taxonomies"
<u>size1_id</u>	Aggregate size class, see table 8.45 WRB 4th edition 2022. See taxonomy "AGGREGATE_SIZES" in the manual "S4M Taxonomies"
<u>size2_id</u>	Aggregate size class, see table 8.45 WRB 4th edition 2022. See taxonomy "AGGREGATE_SIZES" in the manual "S4M Taxonomies"
<u>abundance1</u>	Abundance of size class 1 by volume [%]
<u>abundance2</u>	Abundance of size class 2 by volume [%]

Layer non matrix pore

Layer non-matrix pore Info Section for Soil Point Data

Legend



SIS code name: layer_non_matrix_pore_geo

Short abstract: Layer non-matrix pore Info Section for Soil Point Data

Category: Point Soil Data

Indicator table fields:

name	description\
<u>id</u>	Identifier of the record and Identifier of the point soil data
<u>point_id</u>	Identifier of the point in Point General Data
<u>point_type</u>	Point Soil Data type See taxonomy "POINTS_DATA_TYPES" in the manual "S4M Taxonomies"
<u>date</u>	Survey date associated with the point
<u>project</u>	Identifier or name for the project, or survey activity/campaign, in the frame of which the point soil data data were generated
<u>layer_id</u>	Identifier of the record in Layer General Data
<u>upper</u>	Layer upper depth
<u>lower</u>	Layer lower depth
<u>survey_m_id</u>	Method used to carry out the field survey in order to allow correct interpretation of the information entered. See taxonomy "SURVEY_METHODS" in the manual "S4M Taxonomies"
<u>type1_id</u>	Type 1 of non-matrix pores, see table 8.46 WRB 4th edition 2022 modified by adding the value 'other' in the type field. See taxonomy "NONMATRIX_PORES_TYPES" in the manual "S4M Taxonomies"
<u>size1_id</u>	Dominant pore size class of the non matrix pore type 1, see table 8.47 4th edition 2022. See taxonomy "PORE_SIZES" in

	the manual "S4M Taxonomies"
<u>abund1_id</u>	Abundance of the non matrix pore type 1, see table 8.48 WRB 4th edition 2022. See taxonomy "PORE_ABUNDANCE" in the manual "S4M Taxonomies"
<u>type2_id</u>	Type 2 of non-matrix pores, see table 8.46 WRB 4th edition 2022 modified by adding the value other in the type field. See taxonomy "NONMATRIX_PORES_TYPES" in the manual "S4M Taxonomies"
<u>size2_id</u>	Dominant pore size class of the non matrix pore type 2, see table 8.47 4th edition 2022. See taxonomy "PORE_SIZES" in the manual "S4M Taxonomies"
<u>abund2_id</u>	Abundance of the non matrix pore type 2, see table 8.48 WRB 4th edition 2022. See taxonomy "PORE_ABUNDANCE" in the manual "S4M Taxonomies"
<u>type3_id</u>	Type 3 of non-matrix pores, see table 8.46 WRB 4th edition 2022 modified by adding the value other in the type field. See taxonomy "NONMATRIX_PORES_TYPES" in the manual "S4M Taxonomies"
<u>size3_id</u>	Dominant pore size class of the non matrix pore type 3, see table 8.47 4th edition 2022. See taxonomy "PORE_SIZES" in the manual "S4M Taxonomies"
<u>abund3_id</u>	Abundance of the non matrix pore type 3, see table 8.48 WRB 4th edition 2022. See taxonomy "PORE_ABUNDANCE" in the manual "S4M Taxonomies"
<u>type4_id</u>	Type 4 of non-matrix pores, see table 8.46 WRB 4th edition 2022 modified by adding the value other in the type field. See taxonomy "NONMATRIX_PORES_TYPES" in the manual "S4M Taxonomies"
<u>size4_id</u>	Dominant pore size class of the non matrix pore type 4, see table 8.47 4th edition 2022. See taxonomy "PORE_SIZES" in the manual "S4M Taxonomies"
<u>abund4_id</u>	Abundance of the non matrix pore type 4, see table 8.48 WRB 4th edition 2022. See taxonomy "PORE_ABUNDANCE" in the manual "S4M Taxonomies"

Layer cracks

Layers cracks Info Section for Soil Point Data

Legend



SIS code name: layer_cracks_geo

Short abstract: Layers cracks Info Section for Soil Point Data

Category: Point Soil Data

Indicator table fields:

name	description\
<u>id</u>	Identifier of the record and Identifier of the point soil data
<u>point_id</u>	Identifier of the point in Point General Data
<u>point_type</u>	Point Soil Data type See taxonomy "POINTS_DATA_TYPES" in the manual "S4M Taxonomies"
<u>date</u>	Survey date associated with the point
<u>project</u>	Identifier or name for the project, or survey activity/campaign, in the frame of which the point soil data data were generated
<u>layer_id</u>	Identifier of the record in Layer General Data
<u>upper</u>	Layer upper depth
<u>lower</u>	Layer lower depth
<u>survey_m_id</u>	Method used to carry out the field survey in order to allow correct interpretation of the information entered. See taxonomy "SURVEY_METHODS" in the manual "S4M Taxonomies"
<u>persistence_id</u>	Persistence of cracks, see table 8.49 WRB 4th edition 2022. See taxonomy "PERSISTENCE" in the manual "S4M Taxonomies"
<u>continuity_id</u>	Continuity of cracks, see table 8.50 WRB 4th edition 2022. See taxonomy "CRACKS_CONTINUITY" in the

	manual "S4M Taxonomies"
<u>avg_width</u>	Cracks average width [mm]
<u>abundance</u>	Abundance [number of cracks]

Layer matrix colours

Layers matrix colours Info Section for Soil Point Data

Legend



SIS code name: layer_matrix_colours_geo

Short abstract: Layers matrix colours Info Section for Soil Point Data

Category: Point Soil Data

Indicator table fields:

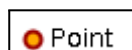
name	description
<u>id</u>	Identifier of the record and Identifier of the point soil data
<u>point_id</u>	Identifier of the point in Point General Data
<u>point_type</u>	Point Soil Data type See taxonomy "POINTS_DATA_TYPES" in the manual "S4M Taxonomies"
<u>date</u>	Survey date associated with the point
<u>project</u>	Identifier or name for the project, or survey activity/campaign, in the frame of which the point soil data data were generated
<u>layer_id</u>	Identifier of the record in Layer General Data
<u>upper</u>	Layer upper depth
<u>lower</u>	Layer lower depth
<u>survey_m_id</u>	Method used to carry out the field survey in order to allow correct interpretation of the information entered. See taxonomy "SURVEY_METHODS" in the manual "S4M Taxonomies"

<u>munsell_m1</u>	Munsell moist dominant colour of the soil matrix
<u>munsell_d1</u>	Munsell dry dominant colour of the soil matrix
<u>area1</u>	Percentage of the exposed area dominant colour
<u>munsell_m2</u>	Munsell moist second colour of the soil matrix
<u>munsell_d2</u>	Munsell dry second colour of the soil matrix
<u>area2</u>	Percentage of the exposed area for colour 2
<u>munsell_m3</u>	Munsell moist third colour of the soil matrix
<u>munsell_d3</u>	Munsell dry third colour of the soil matrix
<u>area3</u>	Percentage of the exposed area for colour 3

Layer Lithogenic Variegates

Layers lithogenic variegates info Section for Soil Point Data

Legend



SIS code name: layer_lithogenic_variegates_geo

Short abstract: Layers lithogenic variegates info Section for Soil Point Data

Category: Point Soil Data

Involved SIS tables: XXXlayer_lithogenic_variegates

Indicator table fields:

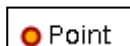
name	description
<u>id</u>	Identifier of the record and Identifier of the point soil data
<u>point_id</u>	Identifier of the point in Point General Data
<u>point_type</u>	Point Soil Data type See taxonomy "POINTS_DATA_TYPES" in the manual "S4M Taxonomies"
<u>date</u>	Survey date associated with the point

<u>project</u>	Identifier or name for the project, or survey activity/campaign, in the frame of which the point soil data data were generated
<u>layer_id</u>	Identifier of the record in Layer General Data
<u>upper</u>	Layer upper depth
<u>lower</u>	Layer lower depth
<u>survey_m_id</u>	Method used to carry out the field survey in order to allow correct interpretation of the information entered. See taxonomy "SURVEY_METHODS" in the manual "S4M Taxonomies"
<u>munsell_m1</u>	Munsell moist first colour according to the Munsell Color Charts
<u>size1_id</u>	Size of lithogenic variegates for colour 1 It indicates the average length of the greatest dimension see table 8.52 WRB 4th edition 2022. See taxonomy "LITHOGENIC_SIZES" in the manual "S4M Taxonomies"
<u>area1</u>	Percentage of abundance by exposed area for colour 1
<u>munsell_m2</u>	Munsell moist second colour according to the Munsell Color Charts
<u>size2_id</u>	Size of lithogenic variegates for colour 2 It indicates the average length of the greatest dimension see table 8.52 WRB 4th edition 2022. See taxonomy "LITHOGENIC_SIZES" in the manual "S4M Taxonomies"
<u>area2</u>	Percentage of abundance by exposed area for colour 2
<u>munsell_m3</u>	Munsell moist third colour according to the Munsell Color Charts
<u>size3_id</u>	Size of lithogenic variegates for colour 3 It indicates the average length of the greatest dimension see table 8.52 WRB 4th edition 2022. See taxonomy "LITHOGENIC_SIZES" in the manual "S4M Taxonomies"
<u>area3</u>	Percentage of abundance by exposed area for colour 3

Layer Redoximorphic Features

Layers redoximorphic features info section for Soil Point Data

Legend



SIS code name: layer_redoximorphic_geo

Short abstract: Layers redoximorphic Info Section for Soil Point Data

Category: Point Soil Data

Indicator table fields:

name	description
<u>id</u>	Identifier of the record and Identifier of the point soil data
<u>point_id</u>	Identifier of the point in Point General Data
<u>point_type</u>	Point Soil Data type See taxonomy "POINTS_DATA_TYPES" in the manual "S4M Taxonomies"
<u>date</u>	Survey date associated with the point
<u>project</u>	Identifier or name for the project, or survey activity/campaign, in the frame of which the point soil data data were generated
<u>layer_id</u>	Identifier of the record in Layer General Data
<u>upper</u>	Layer upper depth
<u>lower</u>	Layer lower depth
<u>survey_m_id</u>	Method used to carry out the field survey in order to allow correct interpretation of the information entered. See taxonomy "SURVEY_METHODS" in the manual "S4M Taxonomies"
<u>clr1_munsell_m</u>	Munsell moist colour 1 (dominant) according to the Munsell Color Charts

<u>clr1_munsell_d</u>	Munsell dry colour 1 (only if reducti- morphic)
<u>clr1_substance_id</u>	Color 1 Substance of oximorphic and reductimorphic features, see tables 8.53 and 8.54 WRB 4th edition 2022. See taxonomy "REDOXIMORPHIC_SUBSTANCES" in the manual "S4M Taxonomies"
<u>clr1_location_id</u>	Color 1 Location of oximorphic and reductimorphic features, see tables 8.55 and 8.56 WRB 4th edition 2022. See taxonomy "REDOXIMORPHIC_LOCATIONS" in the manual "S4M Taxonomies"
<u>clr1_size1_id</u>	Color 1 Size 1 of oximorphic features, It indicates the average length of the greatest dimension see table 8.57 WRB 4th edition 2022. See taxonomy "OXIMORPHIC_SIZES" in the manual "S4M Taxonomies"
<u>clr1_size2_id</u>	Color 1 Size 2 of oximorphic features, It indicates the average length of the greatest dimension see table 8.57 WRB 4th edition 2022. See taxonomy "OXIMORPHIC_SIZES" in the manual "S4M Taxonomies"
<u>clr1_mottles_c_id</u>	Color 1 Mottles contrast, see table 34 FAO 2006. See taxonomy "MOTTLES_CONTRAST" in the manual "S4M Taxonomies"
<u>clr1_mottles_b_id</u>	Color 1 Mottles boundary see table 35 FAO 2006. See taxonomy "MOTTLES_BOUNDARY_TYPES" in the manual "S4M Taxonomies"
<u>clr1_cement_id</u>	Color 1 Consistence of oximorphic features, see table 8.58 WRB 4th edition 2022. See taxonomy "OXIMORPHIC_CEMENTATION" in the manual "S4M Taxonomies"
<u>clr1_area</u>	Color 1 Total abundance of the parts with oximorphic features and the total abundance of the parts with reductimorphic features
<u>clr2_munsell_m</u>	Munsell moist colour 2 according to the Munsell Color Charts
<u>clr2_munsell_d</u>	Munsell dry colour 2 (only if reducti- morphic)
<u>clr2_substance_id</u>	Color 2 Substance of oximorphic and reductimorphic features, see tables 8.53 and 8.54 WRB 4th edition 2022. See taxonomy "REDOXIMORPHIC_SUBSTANCES" in the manual "S4M Taxonomies"
<u>clr2_location_id</u>	Color 2 Location of oximorphic and reductimorphic features, see tables 8.55 and 8.56 WRB 4th edition 2022. See taxonomy "REDOXIMORPHIC_LOCATIONS" in the manual "S4M Taxonomies"
<u>clr2_size1_id</u>	Color 2 Size 1 of oximorphic features It indicates the average length of the greatest dimension see table 8.57 WRB 4th edition 2022. See taxonomy "OXIMORPHIC_SIZES" in the manual "S4M

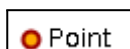
	Taxonomies"
<u>clr2_size2_id</u>	Color 2 Size 2 of oximorphic features It indicates the average length of the greatest dimension see table 8.57 WRB 4th edition 2022. See taxonomy "OXIMORPHIC_SIZES" in the manual "S4M Taxonomies"
<u>clr2_mottles_c_id</u>	Color 2 Mottles contrast, see table 34 FAO 2006. See taxonomy "MOTTLES_CONTRAST" in the manual "S4M Taxonomies"
<u>clr2_mottles_b_id</u>	Color 2 Mottles boundary see table 35 FAO 2006. See taxonomy "MOTTLES_BOUNDARY_TYPES" in the manual "S4M Taxonomies"
<u>clr2_cement_id</u>	Color 2 Consistence of oximorphic features, see table 8.58 WRB 4th edition 2022. See taxonomy "OXIMORPHIC_CEMENTATION" in the manual "S4M Taxonomies"
<u>clr2_area</u>	Color 2 : total abundance of the parts with oximorphic features and the total abundance of the parts with reductimorphic features
<u>clr3_munsell_m</u>	Munsell moist colour 3 according to the Munsell Color Charts
<u>clr3_munsell_d</u>	Munsell dry colour 3 (only if reducti- morphic)
<u>clr3_substance_id</u>	Color 3 Substance of oximorphic and reductimorphic features, see tables 8.53 and 8.54 WRB 4th edition 2022. See taxonomy "REDOXIMORPHIC_SUBSTANCES" in the manual "S4M Taxonomies"
<u>clr3_location_id</u>	Color 3 Location of oximorphic and reductimorphic features, see tables 8.55 and 8.56 WRB 4th edition 2022. See taxonomy "REDOXIMORPHIC_LOCATIONS" in the manual "S4M Taxonomies"
<u>clr3_size1_id</u>	Color 3 Size 1 of oximorphic feature It indicates the average length of the greatest dimension see table 8.57 WRB 4th edition 2022. See taxonomy "OXIMORPHIC_SIZES" in the manual "S4M Taxonomies"
<u>clr3_size2_id</u>	Color 3 Size 2 of oximorphic features It indicates the average length of the greatest dimension see table 8.57 WRB 4th edition 2022. See taxonomy "OXIMORPHIC_SIZES" in the manual "S4M Taxonomies"
<u>clr3_mottles_c_id</u>	Color 3 Mottles contrast, see table 34 FAO 2006. See taxonomy "MOTTLES_CONTRAST" in the manual "S4M Taxonomies"
<u>clr3_mottles_b_id</u>	Color 3 Mottles boundary see table 35 FAO 2006. See taxonomy "MOTTLES_BOUNDARY_TYPES" in the manual "S4M Taxonomies"

<u>clr3_cement_id</u>	Color 3 Consistence of oximorphic features, see table 8.58 WRB 4th edition 2022. See taxonomy "OXIMORPHIC_CEMENTATION" in the manual "S4M Taxonomies"
<u>clr3_area</u>	Color 3 Total abundance of the parts with oximorphic features and the total abundance of the parts with reductimorphic features
<u>oxi_inner</u>	Total abundance of the parts with oximorphic features for inner location
<u>oxi_outer</u>	Total abundance of the parts with oximorphic features for outer location
<u>oxi_random</u>	Total abundance of the parts with oximorphic features for random location
<u>red_inner</u>	Total abundance of the parts with reductimorphic features for inner location
<u>red_outer</u>	Total abundance of the parts with reductimorphic features for outer location
<u>red_random</u>	Total abundance of the parts with reductimorphic features for random location
<u>abund_oxi</u>	Abundance of cemented oximorphic features by volume [%]

Layer coating bridges

Layers coating bridges Info Section for Soil Point Data

Legend



SIS code name: layer_coating_bridges_geo

Short abstract: Layers coating bridges Info Section for Soil Point Data

Category: Point Soil Data

Indicator table fields:

name	description
<u>id</u>	Identifier of the record and Identifier of the point soil data
<u>point_id</u>	Identifier of the point in Point General Data
<u>point_type</u>	Point Soil Data type See taxonomy "POINTS_DATA_TYPES" in the manual "S4M Taxonomies"
<u>date</u>	Survey date associated with the point
<u>project</u>	Identifier or name for the project, or survey activity/campaign, in the frame of which the point soil data data were generated
<u>layer_id</u>	Identifier of the record in Layer General Data
<u>upper</u>	Layer upper depth
<u>lower</u>	Layer lower depth
<u>survey_m_id</u>	Method used to carry out the field survey in order to allow correct interpretation of the information entered. See taxonomy "SURVEY_METHODS" in the manual "S4M Taxonomies"
<u>clay_coat</u>	Abundance of clay coating [%]
<u>form_coat_id</u>	Form of clay coating, see table 14 FAO 2006. See taxonomy "FORM_COATINGS" in the manual "S4M Taxonomies"
<u>clay_bridg_id</u>	Abundance of clay bridges coating [%]
<u>form_bridg_id</u>	Form of clay bridges coating, see table 14 FAO 2006. See taxonomy

	"FORM_COATINGS" in the manual "S4M Taxonomies"
<u>org_coat_id</u>	Form of organic matter coating, see table 14 FAO2006. See taxonomy "ORGANIC_COATINGS" in the manual "S4M Taxonomies"
<u>form_org_id</u>	Form of organic matter coating, see table 14 FAO 2006. See taxonomy "FORM_COATINGS" in the manual "S4M Taxonomies"
<u>crack_coat</u>	Abundance of cracked coatings [%]
<u>sand_silt</u>	Abundance of uncoated sand and coarse silt grains [%]

Layer carbonates

Layers carbonates Info Section for Soil Point Data

Legend



SIS code name: layer_carbonates_geo

Short abstract: Layers carbonates Info Section for Soil Point Data

Category: Point Soil Data

Indicator table fields:

name	description
<u>id</u>	Identifier of the record and Identifier of the point soil data
<u>point_id</u>	Identifier of the point in Point General Data
<u>point_type</u>	Point Soil Data type See taxonomy "POINTS_DATA_TYPES" in the manual "S4M Taxonomies"
<u>date</u>	Survey date associated with the point
<u>project</u>	Identifier or name for the project, or survey activity/campaign, in the frame of which the point soil data data were generated
<u>layer_id</u>	Identifier of the record in Layer General Data
<u>upper</u>	Layer upper depth

<u>lower</u>	Layer lower depth
<u>survey_m_id</u>	Method used to carry out the field survey in order to allow correct interpretation of the information entered. See taxonomy "SURVEY_METHODS" in the manual "S4M Taxonomies"
<u>matr_c_id</u>	Carbonate contents, see table 8.62 WRB 4th edition 2022 See taxonomy "CARBONATE_CONTENTS" in the manual "S4M Taxonomies"
<u>matr_c_ret_id</u>	Retarded reaction with HCl, see table 8.63 WRB 4th edition 2022 See taxonomy "RETARDED_REACTION" in the manual "S4M Taxonomies"
<u>sec_type1_id</u>	type 1 of secondary carbonates table 8.64 WRB 4th edition 2022 See taxonomy "SECONDARY_CARBONATE_TYPES" in the manual "S4M Taxonomies"
<u>sec_type2_id</u>	type 2 of secondary carbonates table 8.64 WRB 4th edition 2022 See taxonomy "SECONDARY_CARBONATE_TYPES" in the manual "S4M Taxonomies"
<u>sec_type3_id</u>	type 3 of secondary carbonates table 8.64 WRB 4th edition 2022 See taxonomy "SECONDARY_CARBONATE_TYPES" in the manual "S4M Taxonomies"
<u>sec_type4_id</u>	type 4 of secondary carbonates table 8.64 WRB 4th edition 2022. See taxonomy "SECONDARY_CARBONATE_TYPES" in the manual "S4M Taxonomies"
<u>sec_size1_id</u>	Mineral size of type 1, see table 15 FAO 2006 See taxonomy "MINERAL_SIZES" in the manual "S4M Taxonomies"
<u>sec_size2_id</u>	Mineral size of type 2, see table 15 FAO 2006 See taxonomy "MINERAL_SIZES" in the manual

	"S4M Taxonomies"
<u>sec_size3_id</u>	Mineral size of type 3, see table 15 FAO 2006 See taxonomy "MINERAL_SIZES" in the manual "S4M Taxonomies"
<u>sec_size4_id</u>	Mineral size of type 4, see table 15 FAO 2006 See taxonomy "MINERAL_SIZES" in the manual "S4M Taxonomies"
<u>sec_shape1_id</u>	Mineral shape of type 1, see table 15 FAO 2006 See taxonomy "MINERAL_SHAPES" in the manual "S4M Taxonomies"
<u>sec_shape2_id</u>	Mineral shape of type 2, see table 15 FAO 2006 See taxonomy "MINERAL_SHAPES" in the manual "S4M Taxonomies"
<u>sec_shape3_id</u>	Mineral shape of type 3, see table 15 FAO 2006 See taxonomy "MINERAL_SHAPES" in the manual "S4M Taxonomies"
<u>sec_shape4_id</u>	Mineral shape of type 4, see table 15 FAO 2006 See taxonomy "MINERAL_SHAPES" in the manual "S4M Taxonomies"
<u>sec_abund1_id</u>	Abundance of secondary carbonate type 1 [%]
<u>sec_abund2_id</u>	Abundance of secondary carbonate type 2 [%]
<u>sec_abund3_id</u>	Abundance of secondary carbonate type 3 [%]
<u>sec_abund4_id</u>	Abundance of secondary carbonate type 4 [%]

Layer gypsum



<u>sgypsum2_id</u>	Gypsum type2 content in the soil matrix (only in layers with little readily soluble salts), see table 16 FAO 2006 See taxonomy "GYPSUM_TYPES" in the manual "S4M Taxonomies"
<u>type1_size_id</u>	Gypsum type1 size, see table 15 FAO 2006 See taxonomy "MINERAL_SIZES" in the manual "S4M Taxonomies"
<u>type2_size_id</u>	Gypsum type2 size, see table 15 FAO 2006 See taxonomy "MINERAL_SIZES" in the manual "S4M Taxonomies"
<u>type1_shape_id</u>	Gypsum type1 shape, see table 15 FAO 2006 See taxonomy "MINERAL_SHAPES" in the manual "S4M Taxonomies"
<u>type2_shape_id</u>	Gypsum type2 shape, see table 15 FAO 2006 See taxonomy "MINERAL_SHAPES" in the manual "S4M Taxonomies"

Layer secondary silica

Layers secondary silica Info Section for Soil Point Data

Legend



SIS code name: layer_secondary_silica_geo

Short abstract: Layers secondary silica Info Section for Soil Point Data

Category: Point Soil Data

Indicator table fields:

name	description
<u>id</u>	Identifier of the record and Identifier of the point soil data
<u>point_id</u>	Identifier of the point in Point General Data
<u>point_type</u>	Point Soil Data type See taxonomy "POINTS_DATA_TYPES" in the manual "S4M Taxonomies"
<u>date</u>	Survey date associated with the point

<u>project</u>	Identifier or name for the project, or survey activity/campaign, in the frame of which the point soil data data were generated
<u>layer_id</u>	Identifier of the record in Layer General Data
<u>upper</u>	Layer upper depth
<u>lower</u>	Layer lower depth
<u>survey_m_id</u>	Method used to carry out the field survey in order to allow correct interpretation of the information entered. See taxonomy "SURVEY_METHODS" in the manual "S4M Taxonomies"
<u>type1_id</u>	Type1 of secondary silica, see table 8.67 4th edition 2022 See taxonomy "SECONDARY_SILICA_TYPES" in the manual "S4M Taxonomies"
<u>type2_id</u>	Type2 of secondary FC), see table 8.68 WRB 4th edition 2022 See taxonomy "SECONDARY_SILICA_TYPES" in the manual "S4M Taxonomies"
<u>dnfcsize1_id</u>	Size class of secondary silica type 1 (only DN and[%] See taxonomy "DNFC_SIZES" in the manual "S4M Taxonomies"
<u>dnfcsize2_id</u>	Size class of secondary silica type 2 (only DN and FC), see table 8.68 WRB 4th edition 2022 See taxonomy "DNFC_SIZES" in the manual "S4M Taxonomies"
<u>abund</u>	Total abundance of secondary silica by exposed area [%]
<u>abund_dnfc</u>	Abundance of DN and FC with size ≥ 1 cm by volume [%]

Layer consistence

Layers consistence Info Section for Soil Point Data

Legend



SIS code name: layer_consistence_geo

Short abstract: Layers consistence Info Section for Soil Point Data

Category: Point Soil Data

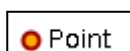
Indicator table fields:

name	description
<u>id</u>	Identifier of the record and Identifier of the point soil data
<u>point_id</u>	Identifier of the point in Point General Data
<u>point_type</u>	Point Soil Data type See taxonomy "POINTS_DATA_TYPES" in the manual "S4M Taxonomies"
<u>date</u>	Survey date associated with the point
<u>project</u>	Identifier or name for the project, or survey activity/campaign, in the frame of which the point soil data data were generated
<u>layer_id</u>	Identifier of the record in Layer General Data
<u>upper</u>	Layer upper depth
<u>lower</u>	Layer lower depth
<u>survey_m_id</u>	Method used to carry out the field survey in order to allow correct interpretation of the information entered. See taxonomy "SURVEY_METHODS" in the manual "S4M Taxonomies"
<u>cement</u>	Cemented consistence by volume [%]
<u>cement_ag1_id</u>	Cementing agent 1, see table 8.35 WRB 4th edition 2022 See taxonomy "CEMENTING_AGENTS" in the manual

	"S4M Taxonomies"
<u>cement_ag2_id</u>	Cementing agent 2, see table 8.35 WRB 4th edition 2022 See taxonomy "CEMENTING_AGENTS" in the manual "S4M Taxonomies"
<u>cement_ag3_id</u>	Cementing agent 3, see table 8.35 WRB 4th edition 2022 See taxonomy "CEMENTING_AGENTS" in the manual "S4M Taxonomies"
<u>cement_cls_id</u>	Cementation class, see table 8.73 WRB 4th edition 2022 See taxonomy "CEMENTATION_CLASSES" in the manual "S4M Taxonomies"
<u>rrclass_m_id</u>	Moist rupture resistance class, see table 8.74 WRB 4th edition 2022 See taxonomy "RRCLASSES_MOIST" in the manual "S4M Taxonomies"
<u>rrclass_d_id</u>	Dry rupture resistance class, see table 8.74 WRB 4th edition 2022 See taxonomy "RRCLASSES_DRY" in the manual "S4M Taxonomies"
<u>susceptib_id</u>	Susceptibility for cementation, see table 8.75 WRB 4th edition 2022 See taxonomy "SUSCEPTIBILITY" in the manual "S4M Taxonomies"
<u>m_failure_id</u>	Type of manner of failure (brittleness), see table 8.76 WRB 4th edition 2022 See taxonomy "MANNER_FAILURE" in the manual "S4M Taxonomies"
<u>plastic_id</u>	Type of plasticity, see table 8.77 WRB 4th edition 2022 See taxonomy "PLASTICITY" in the manual "S4M Taxonomies"
<u>penet_res_id</u>	Penetration resistance
<u>stickiness_id</u>	Stickiness, see table 55 FAO 2006 See taxonomy "STICKINESS" in the manual "S4M Taxonomies"

Layer permafrost

Legend



SIS code name: layer_permafrost_geo

Short abstract: Layers permafrost Info Section for Soil Point Data

Category: Point Soil Data

Indicator table fields:

name	description
<u>id</u>	Identifier of the record and Identifier of the point soil data
<u>point_id</u>	Identifier of the point in Point General Data
<u>point_type</u>	Point Soil Data type See taxonomy "POINTS_DATA_TYPES" in the manual "S4M Taxonomies"
<u>date</u>	Survey date associated with the point
<u>project</u>	Identifier or name for the project, or survey activity/campaign, in the frame of which the point soil data data were generated
<u>layer_id</u>	Identifier of the record in Layer General Data
<u>upper</u>	Layer upper depth
<u>lower</u>	Layer lower depth
<u>survey_m_id</u>	Method used to carry out the field survey in order to allow correct interpretation of the information entered. See taxonomy "SURVEY_METHODS" in the manual "S4M Taxonomies"
<u>cry_alter1_id</u>	Cryogenic alteration (feature 1), see table 8.81 WRB 4th edition 2022. See taxonomy "CRYOGENIC_ALTERATION" in the manual "S4M Taxonomies"
<u>cry_alter2_id</u>	Cryogenic alteration (feature 2), see table 8.81 WRB 4th edition 2022. See taxonomy "CRYOGENIC_ALTERATION" in the

	manual "S4M Taxonomies"
<u>cry_alter3_id</u>	Cryogenic alteration (feature 3), see table 8.81 WRB 4th edition 2022 See taxonomy "CRYOGENIC_ALTERATION" in the manual "S4M Taxonomies"
<u>cry_abund1</u>	Abundance (feature 1) by exposed area [%]
<u>cry_abund2</u>	Abundance (feature 2) by exposed area [%]
<u>cry_abund3</u>	Abundance (feature 3) by exposed area [%]
<u>permafrost_id</u>	Cryogenic alteration, see table 8.81 WRB 4th edition 2022 See taxonomy "PERMAFROST_TYPES" in the manual "S4M Taxonomies"

Layer organic carbon

Legend



SIS code name: XXX

Short abstract: Layers organic carbon Info Section for Soil Point Data

Category: Point Soil Data

Involved SIS tables: XXXlayer_organic_carbon

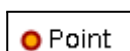
Indicator table fields:

name	description
<u>id</u>	Identifier of the record and Identifier of the point soil data
<u>point_id</u>	Identifier of the point in Point General Data
<u>point_type</u>	Point Soil Data type See taxonomy "POINTS_DATA_TYPES" in the manual "S4M Taxonomies"
<u>date</u>	Survey date associated with the point
<u>project</u>	Identifier or name for the project, or survey activity/campaign, in the frame of which the point soil data data were generated
<u>layer_id</u>	Identifier of the record in Layer General Data
<u>upper</u>	Layer upper depth

<u>lower</u>	Layer lower depth
<u>survey_m_id</u>	Method used to carry out the field survey in order to allow correct interpretation of the information entered. See taxonomy "SURVEY_METHODS" in the manual "S4M Taxonomies"
<u>contentmin</u>	Minimum content of organic carbon [%]
<u>contentmax</u>	Maximum content of organic carbon [%]
<u>nat_accum1_id</u>	Type 1 of accumulation of organic matter, see table 8.85 WRB 4th edition 2022 See taxonomy "ACCUMULATION_TYPES" in the manual "S4M Taxonomies"
<u>nat_accum2_id</u>	Type 2 of accumulation of organic matter, see table 8.85 WRB 4th edition 2022 See taxonomy "ACCUMULATION_TYPES" in the manual "S4M Taxonomies"
<u>nat_accum3_id</u>	Type 3 of accumulation of organic matter, see table 8.85 WRB 4th edition 2022 See taxonomy "ACCUMULATION_TYPES" in the manual "S4M Taxonomies"
<u>abundance1</u>	Abundance of type 1 by exposed area [%]
<u>abundance2</u>	Abundance of type 2 by exposed area [%]
<u>abundance3</u>	Abundance of type 3 by exposed area [%]
<u>black_carb</u>	Black carbon by exposed area [%]

Layer roots

Legend



SIS code name: layer_roots_geo

Short abstract: Layers roots Info Section for Soil Point Data

Category: Point Soil Data

Indicator table fields:

name	description
<u>id</u>	Identifier of the record and Identifier of the point soil data
<u>point_id</u>	Identifier of the point in Point General Data
<u>point_type</u>	Point Soil Data type See taxonomy "POINTS_DATA_TYPES" in the manual "S4M Taxonomies"
<u>date</u>	Survey date associated with the point
<u>project</u>	Identifier or name for the project, or survey activity/campaign, in the frame of which the point soil data data were generated
<u>layer_id</u>	Identifier of the record in Layer General Data
<u>upper</u>	Layer upper depth
<u>lower</u>	Layer lower depth
<u>survey_m_id</u>	Method used to carry out the field survey in order to allow correct interpretation of the information entered. See taxonomy "SURVEY_METHODS" in the manual "S4M Taxonomies"
<u>a_lt2mm_id</u>	Abundance of roots ≤ 2 mm see table 8.86 WRB 4th edition 2022 modified by inserting 2 classes: 0.2-2mm and 2-5mm. See taxonomy "ROOT_ABUNDANCE_CLASSES" in the manual "S4M Taxonomies"
<u>a_lt05mm_id</u>	Abundance of very fine roots (<0.5 mm) see table 8.86 WRB 4th edition 2022 modified by inserting 2 classes: 0.2-2mm and 2-5mm See taxonomy "ROOT_ABUNDANCE_CLASSES" in the manual

	"S4M Taxonomies"
<u>a_05to2mm_id</u>	Abundance of very fine roots (0.5-2 mm) see table 8.86 WRB 4th edition 2022 modified by inserting 2 classes: 0.2-2mm and 2-5mm See taxonomy "ROOT_ABUNDANCE_CLASSES" in the manual "S4M Taxonomies"
<u>a_gt2mm_id</u>	Abundance of roots > 2 mm see table 8.86 WRB 4th edition 2022 modified by inserting 2 classes: 0.2-2mm and 2-5mm See taxonomy "ROOT_ABUNDANCE_CLASSES" in the manual "S4M Taxonomies"
<u>a_2to5mm_id</u>	Abundance of very fine roots (2-5 mm) see table 8.86 WRB 4th edition 2022 modified by inserting 2 classes: 0.2-2mm and 2-5mm See taxonomy "ROOT_ABUNDANCE_CLASSES" in the manual "S4M Taxonomies"
<u>a_gt5mm_id</u>	Abundance of coarse roots roots > 5 mm see table 8.86 WRB 4th edition 2022 modified by inserting 2 classes: 0.2-2mm and 2-5mm See taxonomy "ROOT_ABUNDANCE_CLASSES" in the manual "S4M Taxonomies"

Layer animal activity

Legend



SIS code name: layer_animal_activities_geo

Short abstract: Layers animal activities Info Section for Soil Point Data

Category: Point Soil Data

Indicator table fields:

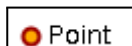
name	description
<u>id</u>	Identifier of the record and Identifier of the point soil data
<u>point_id</u>	Identifier of the point in Point General Data
<u>point_type</u>	Point Soil Data type See taxonomy "POINTS_DATA_TYPES" in the manual "S4M Taxonomies"
<u>date</u>	Survey date associated with the point

<u>project</u>	Identifier or name for the project, or survey activity/campaign, in the frame of which the point soil data data were generated
<u>layer_id</u>	Identifier of the record in Layer General Data
<u>upper</u>	Layer upper depth
<u>lower</u>	Layer lower depth
<u>survey_m_id</u>	Method used to carry out the field survey in order to allow correct interpretation of the information entered. See taxonomy "SURVEY_METHODS" in the manual "S4M Taxonomies"
<u>type1_id</u>	Type 1 of animal activity see table 8.87 WRB 4th edition 2022 See taxonomy "ANIMAL_ACTIVITY_TYPES" in the manual "S4M Taxonomies"
<u>type2_id</u>	Type 2 of animal activity see table 8.87 WRB 4th edition 2022 See taxonomy "ANIMAL_ACTIVITY_TYPES" in the manual "S4M Taxonomies"
<u>type3_id</u>	Type 3 of animal activity see table 8.87 WRB 4th edition 2022 See taxonomy "ANIMAL_ACTIVITY_TYPES" in the manual "S4M Taxonomies"
<u>type4_id</u>	Type 4 of animal activity see table 8.87 WRB 4th edition 2022 See taxonomy "ANIMAL_ACTIVITY_TYPES" in the manual "S4M Taxonomies"
<u>type5_id</u>	Type 5 of animal activity see table 8.87 WRB 4th edition 2022 See taxonomy "ANIMAL_ACTIVITY_TYPES" in the manual "S4M Taxonomies"
<u>mammal</u>	Mammal activity by exposed area [%]
<u>bird</u>	Bird activity by exposed area [%]
<u>worm</u>	Worm activity by exposed area silica, see table 8.67 4th edition 2022
<u>insect</u>	Insect activity by exposed area [%]
<u>unspecify</u>	Unspecified activity by exposed area [%]

Layer Human Alterations

Layers human alterations Info Section for Soil Point Data

Legend



SIS code name: layer_human_alterations_geo

Short abstract: Layers human alterations Info Section for Soil Point Data

Category: Point Soil Data

Indicator table fields:

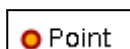
name	description
<u>id</u>	Identifier of the record and Identifier of the point soil data
<u>point_id</u>	Identifier of the point in Point General Data
<u>point_type</u>	Point Soil Data type See taxonomy "POINTS_DATA_TYPES" in the manual "S4M Taxonomies"
<u>date</u>	Survey date associated with the point
<u>project</u>	Identifier or name for the project, or survey activity/campaign, in the frame of which the point soil data data were generated
<u>layer_id</u>	Identifier of the record in Layer General Data
<u>upper</u>	Layer upper depth
<u>lower</u>	Layer lower depth
<u>survey_m_id</u>	Method used to carry out the field survey in order to allow correct interpretation of the information entered. See taxonomy "SURVEY_METHODS" in the manual "S4M Taxonomies"
<u>nat_mat1_id</u>	Artificial additions of natural material 1 see table 8.88 WRB 4th edition 2022 See taxonomy "NATURAL_MATERIAL_TYPES" in the

	manual "S4M Taxonomies"
<u>nat_mat2_id</u>	Artificial additions of natural material 2 see table 8.88 WRB 4th edition 2022. See taxonomy "NATURAL_MATERIAL_TYPES" in the manual "S4M Taxonomies"
<u>nat_mat3_id</u>	Artificial additions of natural material 3 see table 8.88 WRB 4th edition 2022 See taxonomy "NATURAL_MATERIAL_TYPES" in the manual "S4M Taxonomies"
<u>abundance1</u>	Abundance of material 1 by volume [%]
<u>abundance2</u>	Abundance of material 2 by volume [%]
<u>abundance3</u>	Abundance of material 3 by volume [%]
<u>texture_id</u>	Texture class, see table 8.89 WRB 4th edition 2022 See taxonomy "TEXTURE_CLASSES" in the manual "S4M Taxonomies"
<u>carbonate_id</u>	Carbonate content, see table 8.64 WRB 4th edition 2022 See taxonomy "SECONDARY_CARBONATE_TYPES" in the manual "S4M Taxonomies"
<u>carbon</u>	Soil organic carbon content see table 8.4.36 WRB 4th edition 2022
<u>alter1_id</u>	In situ primary alteration 1 see table 8.89 WRB 4th edition 2022 See taxonomy "HUMAN_ALTERATIONS_TYPES" in the manual "S4M Taxonomies"
<u>alter2_id</u>	In situ secondary alteration see table 8.89 WRB 4th edition 2022 See taxonomy "HUMAN_ALTERATIONS_TYPES" in the manual "S4M Taxonomies"
<u>aggregate_id</u>	Soil aggregate formation after additions or after in-situ alterations, see table 8.90 WRB 4th edition 2022 See taxonomy "AGGREGATE_FORMATION" in the manual "S4M Taxonomies"

Layer degree decomposition

Layers degree decomposition Info Section for Soil Point Data

Legend



SIS code name: layer_degree_decomposition_geo

Short abstract: Layers degree decomposition Info Section for Soil Point Data

Category: Point Soil Data

Indicator table fields:

name	description
<u>id</u>	Identifier of the record and Identifier of the point soil data
<u>point_id</u>	Identifier of the point in Point General Data
<u>point_type</u>	Point Soil Data type See taxonomy "POINTS_DATA_TYPES" in the manual "S4M Taxonomies"
<u>date</u>	Survey date associated with the point
<u>project</u>	Identifier or name for the project, or survey activity/campaign, in the frame of which the point soil data data were generated
<u>layer_id</u>	Identifier of the record in Layer General Data
<u>upper</u>	Layer upper depth
<u>lower</u>	Layer lower depth
<u>survey_m_id</u>	Method used to carry out the field survey in order to allow correct interpretation of the information entered. See taxonomy "SURVEY_METHODS" in the manual "S4M Taxonomies"
<u>vis_plant</u>	Visible plant tissues by volume [%]
<u>sbdiv_horz_id</u>	Subdivisions of the Oa horizon, see table 8.92 WRB 4th edition 2022. See taxonomy "SUBDIVISION_HORIZON" in the manual "S4M Taxonomies"
<u>plant_res1_id</u>	Dead residues 1 of specific plants see table 8.93 WRB 4th edition 2022. See taxonomy "DEAD_PLANT_TYPES" in the

	manual "S4M Taxonomies"
<u>plant_res2_id</u>	Dead residues 2 of specific plants see table 8.93 WRB 4th edition 2022. See taxonomy "DEAD_PLANT_TYPES" in the manual "S4M Taxonomies"
<u>abundance1</u>	Abundance of dead natural plant residues 1 by volume [%]
<u>abundance2</u>	Abundance of dead natural plant residues 2 by volume [%]

Laboratory Data section

Laboratory data

Laboratory Data Info Section for Soil Point Data

Legend



SIS code name: lab_data_geo

Short abstract: Laboratory Data Info Section for Soil Point Data

Category: Point Soil Data

Indicator table fields:

name	description
<u>id</u>	Identifier of the record and Identifier of the point soil data
<u>point_id</u>	Identifier of the point in Point General Data
<u>point_type</u>	Point Soil Data type See taxonomy "POINTS_DATA_TYPES" in the manual "S4M Taxonomies"
<u>date</u>	Survey date associated with the point
<u>project</u>	Identifier or name for the project, or survey activity/campaign, in the frame of which the point soil data data were generated
<u>l_number</u>	sequence number of the layer in the profile, can be null

<u>horizon</u>	horizon designation of the layer in the profile, can be null
<u>upper</u>	Layer upper depth
<u>lower</u>	Layer lower depth
<u>survey_m_id</u>	Method used to carry out the field survey in order to allow correct interpretation of the information entered. See taxonomy "SURVEY_METHODS" in the manual "S4M Taxonomies"
<u>gravel</u>	Gravel content (%)
<u>cls_sys_id</u>	Classification system used for texture of fine earth See taxonomy "G_CLASSIFICATION_SYSTEMS" in the manual "S4M Taxonomies"
<u>sand</u>	Sand (% of the fine earth)
<u>v_c_sand</u>	Very coarse sand (% of the fine earth)
<u>c_sand</u>	Coarse sand (% of the fine earth)
<u>m_sand</u>	Medium sand (% of the fine earth)
<u>f_sand</u>	Fine sand (% of the fine earth)
<u>v_f_sand</u>	Very fine sand (% of the fine earth)
<u>met_sand_id</u>	Method used for sand content See taxonomy "SAND_CONTENT_METHODS" in the manual "S4M Taxonomies"
<u>silt</u>	Silt (% of the fine earth)
<u>c_silt</u>	Coarse silt (% of the fine earth)
<u>f_silt</u>	Fine silt (% of the fine earth)
<u>met_silt_id</u>	Method used for silt content See taxonomy "SILT_CLAY_CONTENT_METHODS" in the manual "S4M Taxonomies"
<u>clay</u>	Clay (% of the fine earth)
<u>met_clay_id</u>	Method used for clay content See taxonomy "SILT_CLAY_CONTENT_METHODS" in the manual "S4M Taxonomies"
<u>texture_id</u>	Texture class See taxonomy "TEXTURE_CLASSES" in the manual "S4M Taxonomies"
<u>bulk_dens</u>	Bulk density (g/cm3)
<u>met_bulk_dens_id</u>	Method used for Bulk Density See taxonomy "BULK_DENSITY_METHODS" in the manual "S4M Taxonomies"
<u>slake_test</u>	Slake test
<u>met_slake_test</u>	Method used for Slake test See taxonomy "XXX" in the manual "S4M Taxonomies"
<u>el_cond</u>	Electrical conductivity (dS/m)
<u>met_el_cond_id</u>	Laboratory method used to evaluate Electrical conductivity

	See taxonomy "EL_CONDUCTIVITY_PH_METHODS" in the manual "S4M Taxonomies"
<u>hy_cond</u>	Hydraulic conductivity at saturation (mm/h)
<u>met_hy_cond_id</u>	Method used for Hydraulic conductivity at saturation See taxonomy "HYDRAULIC_CONDUCTIVITY_METHODS" in the manual "S4M Taxonomies"
<u>satur</u>	Water Saturation (%)
<u>field_cap</u>	Field capacity (%)
<u>wilting_p</u>	Wilting point (%)
<u>awc</u>	AWC (%)
<u>met_s_f_w_id</u>	Method used for saturation, field capacity and wilting point See taxonomy "WILTING_POINT_METHODS" in the manual "S4M Taxonomies"
<u>acidity</u>	Exchangeable acidity (cmol/kg)
<u>met_acidity</u>	Method used for Exchangeable acidity
<u>ph_h2o</u>	pH (H2O)
<u>met_ph_h2o_id</u>	Method used for pH (H2O) See taxonomy "EL_CONDUCTIVITY_PH_METHODS" in the manual "S4M Taxonomies"
<u>ph_kcl</u>	pH (KCl)
<u>met_ph_kcl_id</u>	Method used for pH (KCl) See taxonomy "EL_CONDUCTIVITY_PH_METHODS" in the manual "S4M Taxonomies"
<u>ph_ccl</u>	pH (CaCl2)
<u>met_ph_ccl_id</u>	Method used for pH (CaCl2) See taxonomy "EL_CONDUCTIVITY_PH_METHODS" in the manual "S4M Taxonomies"
<u>org_car</u>	Organic Carbon content (g/kg)
<u>met_org_car_id</u>	Method used for Organic Carbon content See taxonomy "ORGANIC_CARBON_METHODS" in the manual "S4M Taxonomies"
<u>org_mat</u>	Organic matter content (%)
<u>met_org_mat_id</u>	Method used for organic matter content See taxonomy "ORGANIC_MATTER_CONTENT_METHODS" in the manual "S4M Taxonomies"
<u>caco3_content</u>	Calcium carbonate CaCO3 content (%)
<u>met_content_caco3_id</u>	Method used for Calcium carbonate CaCO3 content See taxonomy "CACO3_CONTENT_METHODS" in the manual "S4M Taxonomies"
<u>active_caco3</u>	Active Calcium carbonate CaCO3 (%)
<u>met_active_caco3_id</u>	Laboratory methods used for active Calcium carbonate CaCO3 evaluation See taxonomy "ACTIVE_CACO3_CONTENT_METHODS" in the manual "S4M Taxonomies"
<u>gypsum</u>	Gypsum content (%)

<u>met_gypsum_id</u>	Method used for Gypsum content See taxonomy "GYPSUM_CONTENT_METHODS" in the manual "S4M Taxonomies"
<u>cec</u>	CEC (cmol/Kg)
<u>met_cec_id</u>	Method used for CEC. See taxonomy "CEC_CA_MG_NA_K_METHODS" in the manual "S4M Taxonomies"
<u>ca</u>	Exchangeable Ca++ (cmol/Kg)
<u>mg</u>	Exchangeable Mg++ (cmol/Kg)
<u>na</u>	Exchangeable Na++ (cmol/Kg)
<u>k</u>	Exchangeable K+ (cmol/Kg)
<u>met_exc_id</u>	Laboratory method used to evaluate exchangeable cations. See taxonomy "EXCHANGEABLE_CATIONS_METHODS" in the manual "S4M Taxonomies"
<u>base_saturation</u>	Base saturation (%)
<u>esp</u>	ESP (ratio)
<u>sol_ca</u>	Sol. Ca++ (cmol/L)
<u>sol_mg</u>	Sol. Mg++ (cmol/L)
<u>sol_na</u>	Sol. Na++ (cmol/L)
<u>met_sol_cations_id</u>	Method used for soluble cations See taxonomy "SOLUBLE_CATIONS_METHODS" in the manual "S4M Taxonomies"
<u>sar</u>	SAR (ratio)
<u>n_tot</u>	N total content (g/Kg)
<u>met_n_tot_id</u>	Method used for N tot content See taxonomy "N_CONTENT_METHODS" in the manual "S4M Taxonomies"
<u>p_cont</u>	Available P content (mg/kg)
<u>met_p_cont_id</u>	Method used for available P content See taxonomy "P_CONTENT_METHODS" in the manual "S4M Taxonomies"
<u>nh4</u>	NH4 (mg/kg)
<u>met_nh4_id</u>	Method used for NH4 See taxonomy "NH4_METHODS" in the manual "S4M Taxonomies"
<u>no3</u>	NO3 (mg/kg)
<u>met_no3_id</u>	Method used for NO3 See taxonomy "NO3_METHODS" in the manual "S4M Taxonomies"
<u>roc</u>	ROC is the Remaining Oxidable Carbon also called Elementary Carbon (mg/Kg)
<u>toc400</u>	TOC 400 is organically bound Carbon released up to 400 degrees (g/kg)
<u>met_roc_toc400_id</u>	Method used for ROC and TOC400 See taxonomy "ROC_TOC400_METHODS" in the manual "S4M Taxonomies"

<u>feox</u>	Fe-ox (g/kg)
<u>fed</u>	Fe-d (g/kg)
<u>fep</u>	Fe-p (g/kg)
<u>fe_tot</u>	Total Iron Fe (g/kg)
<u>met_fe_tot_id</u>	Method for Fe tot See taxonomy "FE_TOT_METHODS" in the manual "S4M Taxonomies"
<u>mn</u>	Manganese Mn (mg/kg)
<u>met_mn_id</u>	Method used for Mn See taxonomy "MN_ZN_CU_METHODS" in the manual "S4M Taxonomies"
<u>zn</u>	Zinc Zn (mg/kg)
<u>met_zn_id</u>	Method used for Zn See taxonomy "MN_ZN_CU_METHODS" in the manual "S4M Taxonomies"
<u>cu</u>	Copper Cu (mg/kg)
<u>met_cu_id</u>	Method used for Cu See taxonomy "MN_ZN_CU_METHODS" in the manual "S4M Taxonomies"
<u>pb</u>	Lead Pb (mg/kg)
<u>met_pb_id</u>	Method used for Pb See taxonomy "PB_HG_CD_NI_CR_AS_CO_V_METHODS" in the manual "S4M Taxonomies"
<u>hg</u>	Mercury Hg (mg/kg)
<u>met_hg_id</u>	Method used for Hg See taxonomy "PB_HG_CD_NI_CR_AS_CO_V_METHODS" in the manual "S4M Taxonomies"
<u>cd</u>	Cadmium Cd (mg/kg)
<u>met_cd_id</u>	Method used for Cd See taxonomy "PB_HG_CD_NI_CR_AS_CO_V_METHODS" in the manual "S4M Taxonomies"
<u>ni</u>	Nickel Ni (mg/kg)
<u>met_ni_id</u>	Method used for Ni See taxonomy "PB_HG_CD_NI_CR_AS_CO_V_METHODS" in the manual "S4M Taxonomies"
<u>sb</u>	Antimony Sb (mg/kg)
<u>met_sb_id</u>	Method used for Sb See taxonomy "SB_METHODS" in the manual "S4M Taxonomies"
<u>cr</u>	Chromium Cr (mg/kg)
<u>met_cr_id</u>	Method used for Cr See taxonomy "PB_HG_CD_NI_CR_AS_CO_V_METHODS" in the manual "S4M Taxonomies"
<u>as_value</u>	Arsenic As (mg/kg)
<u>met_as_id</u>	Method used for As

	See taxonomy "PB_HG_CD_NI_CR_AS_CO_V_METHODS" in the manual "S4M Taxonomies"
<u>co</u>	Cobalt Co (mg/kg)
<u>met_co_id</u>	Method used for Co See taxonomy "PB_HG_CD_NI_CR_AS_CO_V_METHODS" in the manual "S4M Taxonomies"
<u>v</u>	Vanadium V (mg/kg)
<u>met_v_id</u>	Method used for V See taxonomy "PB_HG_CD_NI_CR_AS_CO_V_METHODS" in the manual "S4M Taxonomies"
<u>notes</u>	Notes